

The Start of a Great Project

CONTENTS

INTRODUCTION TO THE BOOK	iii
o Mathematical Foundations	
1 Basic Mathematics	3
1 Basics of Isogeny-based Cryptography	
I ELLIPTIC CURVES	9
1 Elliptic Curves (mathematics)	9
2 Elliptic Curves (cryptography)	15
II ISOGENIES	17
III PAIRINGS	19
IV ABELIAN VARIETIES	21
V HIGHER-DIMENSIONAL ISOGENIES	23
2 Advanced Topics in Isogeny-based Cryptography	
3 Cryptographic Constructions	

INTRODUCTION TO THE BOOK

The field of isogeny-based cryptography is rapidly expanding since the 2020s. To keep up with developments in isogeny-based cryptography, we need a strong understanding of the basics, and a gentle introduction to advanced topics. Unfortunately, there is no unified place yet, where anyone can go to learn these topics, nor is there unified notation or sometimes even definitions between authors and works.

This collaborative open-source text book aims to fill that gap by providing introductory materials to all topics in isogeny-based cryptography. We strongly focus on intuition and examples to grasp these materials, to ensure that we turn mathematical insight into concrete cryptographic applications.

We strive to describe every aspect of isogeny-based cryptography, with ample references to all related work that introduced or expanded on these concepts. It will be impossible to keep up with the current progress in isogeny-based cryptography; whenever we have not yet adequately described a topic, we simply list the literature to provide a starting point for the reader.

Part o

MATHEMATICAL FOUNDATIONS

MATHEMATICAL FOUNDATIONS

This part describes the mathematical foundations required to understand the other parts. This contains an overview of the necessary number theory, algebra, or other subjects.

§1 BASIC MATHEMATICS

TODO: add resultant computation

Part 1

BASICS OF ISOGENY-BASED CRYPTOGRAPHY

BASICS OF ISOGENY-BASED CRYPTOGRAPHY

This part describes basic topics in isogeny-based cryptography, such as elliptic curves, isogenies, pairings, and the class group action.

1. [Chapter I: Elliptic curves, curve models, torsion subgroups, arithmetic.](#)
2. [Chapter II: Elliptic Curves \(cryptography\)](#) Isogenies, Velu's formulas, isogeny graphs, endomorphisms, and representations.
3. [Chapter III: Elliptic Curves \(cryptography\)](#) Pairings, the Weil pairing, the Tate pairing, Miller's algorithm, and cubical arithmetic.
4. [Chapter IV: Elliptic Curves \(cryptography\)](#) Abelian varieties, Jacobians of hyperelliptic curves, Mumford representation, and genus 2.
5. [Chapter V: Elliptic Curves \(cryptography\)](#) Higher-dimensional isogenies, Kani's lemma. **TODO: test**

ELLIPTIC CURVES

We start with mathematical aspects of elliptic curves, before we discuss cryptographic aspects and applications.

TODO: small todo list:

- discuss twist
- fix Cref definition theorem mixup

§1 ELLIPTIC CURVES (MATHEMATICS)

The definition of an elliptic curve is the most central definition in isogeny-based cryptography.

Definition 1. An *elliptic curve* over a field k is a pair $(E, \mathbf{0})$, where E is a smooth projective curve of genus 1, and $\mathbf{0} \in E(k)$ a rational point. The *rational points* on an elliptic curve is the set $E(k)$, and for any field extension K/k , we similarly denote by $E(K)$ the K -rational points of E . $\circ$

This definition ensures that an elliptic curve has a group operation on the set $E(K)$ for any K/k . Unfortunately, this definition is almost entirely useless for concrete computations in this group. In practice, we are often given a curve in a given *model*, such as a *short Weierstrass curve*

$$E_{a,b} : y^2 = x^3 + ax + b \quad a, b \in k, \quad (1)$$

a *Montgomery curve*

$$E_{A,B} : By^2 = x^3 + Ax^2 + x, \quad A, B \in k, \quad (2)$$

or perhaps even a *twisted Edwards curve*

$$E_{a,d} : ax^2 + y^2 = 1 + dx^2y^2, \quad a, d \in k. \quad (3)$$

So why does the abstract definition differ so much from the curve representations we use in practice? First, the abstract definition captures all of the curve models above, and resolves some difficulties that we encounter with the *affine* curve models and the mysterious ‘point at infinity’ $\mathbf{0} \in E(k)$. Second, each curve model above is mostly used in specific situations, because not every elliptic curve is isomorphic to a curve in a given curve model, and because specific curve models have arithmetical properties that make them uniquely suitable to certain applications. This last point is crucial once we get to the cryptographical aspects of elliptic curves.

VISUALISING ELLIPTIC CURVES. Having concrete equations helps us in visualising elliptic curves, which gives us the images we are so familiar with. For example, we may draw the curve $E : y^2 = x^3 - 2x + 4$ as follows.

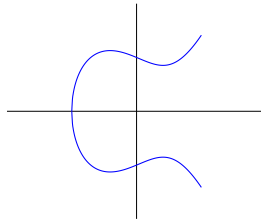


Figure 1: A visualisation over $\mathbb{R}$ of the elliptic curve $y^2 = x^3 - 2x + 4$.

Note: the picture we have drawn here displays the set of points (x, y) that satisfy $y^2 = x^3 - 2x + 4$ where $x, y \in \mathbb{R}$, so we have drawn $E(\mathbb{R})$! Through magic, these visualisations often help our intuition for curves over finite fields too. But we should always be careful that the nature of elliptic curves over $\mathbb{R}$ is very different from the nature of curves over a finite field. In short: we may borrow inspiration from such visualisations, but they may also misrepresent the actual situation we are working with. As an example, the picture of the above curve E over the finite field $\mathbb{F}_7$ looks as follows.

1.1 PROJECTIVE CURVES

By [Definition 1](#), an elliptic curve is *projective*, but the curve models we have given so far are affine. We do so because affine curves are a bit easier to work with, practically, and are easier to visualise. Nevertheless, we should properly

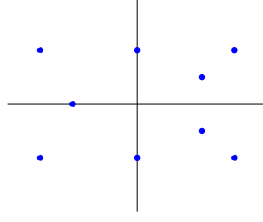


Figure 2: A visualisation over $\mathbb{F}_7$ of the elliptic curve $y^2 = x^3 - 2x + 4$.

define elliptic curves as a projective variety. Luckily, this is not too difficult. We will show this for the affine short Weierstrass curve

$$E_{a,b} : y^2 = x^3 + ax + b.$$

This equation is in fact an affine chart of the ‘proper’ projective elliptic curve

$$\mathbf{E}_{a,b} : Y^2Z = X^3 + aXZ^2 + bZ^3,$$

whose points $P \in \mathbf{E}_{a,b}$ are tuples $(X : Y : Z) \in \mathbb{P}^3(k)$. These points are defined up to scalars, that is, $(X : Y : Z)$ is the same point as $(\lambda X : \lambda Y : \lambda Z)$ for any $\lambda \in k^*$.

Hence, we can simplify the points on $\mathbf{E}_{a,b}$ a bit: for any point $P \in \mathbf{E}_{a,b}(k)$ with $Z \neq 0$, we may normalize $(X : Y : Z)$ as $(\frac{X}{Z} : \frac{Y}{Z} : 1)$. But this then defines a point $(\frac{X}{Z}, \frac{Y}{Z})$ on the affine form of this curve $E_{a,b}(k)$, as it satisfies the curve equation. Vice versa, any point $(x, y) \in E_{a,b}(k)$ defines a point $(x : y : 1) \in \mathbf{E}_{a,b}(k)$.

The only difference between the points in $E_{a,b}(k)$ and $\mathbf{E}_{a,b}(k)$ must therefore be the points on $\mathbf{E}_{a,b}$ with $Z = 0$. But then, the left-hand side of the curve equation resolves to 0, and the right-hand side resolves to X^3 , so we must have $X = 0$ too. This means such points are of the form $(0 : Y : 0)$ with $Y \in k^*$, and these are all equal to $(0 : 1 : 0)$ when scaled by Y^{-1} . Thus, there is only the single point $(0 : 1 : 0)$ that is well-defined for $\mathbf{E}_{a,b}$ but cannot be written as an affine point $(x, y) \in E_{a,b}$. This is the ‘point at infinity’ for $E_{a,b}$, which we denote $\mathbf{0}_E$.

IN PRACTICE. In practice, we may often simply use the affine curve $E_{a,b}$. Whenever a situation is unclear and involves $\mathbf{0}_E$, we will need to switch to the projective view $\mathbf{E}_{a,b}$, which gives us the complete correct view on $E_{a,b}$.

1.2 SMOOTH CURVES

By [Definition 1](#), an elliptic curve is *smooth*, but what does that mean, practically speaking? We use our intuition over $\mathbb{R}$: the curve $C : y^2 = x^3$ is clearly *not* smooth.

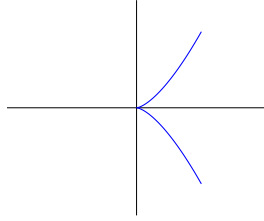


Figure 3: The curve $y^2 = x^3$ is not smooth at $(0,0)$.

The problem is the point $(0,0)$, where C is quite literally very pointy. Although we clearly see this in the geometric picture, the mathematical community has collectively decided that such an approach to ‘smoothness’ is not rigorous enough. Instead, smoothness of a point on a curve can properly be established by non-vanishing differentials, which mathematically captures the ‘pointiness’.

Definition 2. Let C be a curve given by a non-constant polynomial $f(X_1, \dots, X_n) = 0$, then $P \in C$ is a *singular point* when all derivatives vanish, i.e.,

$$\frac{\partial f}{\partial X_1}(P) = 0, \quad \frac{\partial f}{\partial X_2}(P) = 0, \quad \dots, \quad \frac{\partial f}{\partial X_n}(P) = 0.$$

If C has no singular points, we say that C is a *smooth curve*. ◦

This gives us an easy way to verify that (the affine part of) a given curve is indeed smooth. We show this for a curve in short Weierstrass form.

Lemma 3. Let $E_{a,b}$ be a curve in short Weierstrass form $E_{a,b} : y^2 = x^3 + ax + b$. Then $E_{a,b}$ is a smooth curve if and only if $4a^3 + 27b^2 \neq 0$.

Proof. We first show all affine parts are smooth using the curve equation. The point 0_E at infinity can be shown to be smooth using the projective curve equation. We get

$$\frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial x} = -3x^2 - a,$$

and so any potential singular point $P = (x_0, y_0)$ must have $y_0 = 0$. Thus, a singular point would have x_0 as a root of both the right-hand side $x^3 + ax + b$

and its derivative $-\frac{\partial f}{\partial x} = 3x^2 + a$. This only happens when the resultant (see [Section 1](#)) of these two polynomials is zero, and this resultant equals $4a^3 + 27b^2$. $\square$

Similarly, a Montgomery curve $E_{A,B} : By^2 = x^3 + Ax^2 + x$ is a smooth curve when $A^2 \neq 4$ and $B \neq 0$.

1.3 THE SIZE OF ELLIPTIC CURVES

The possible sizes and structures of elliptic curves over a field k are a fascinating subject. We first deal with possible sizes, before we discuss the possible structures. Although the size and structure of an elliptic curve over $E(\mathbb{Q})$, or more generally $E(K)$ for number fields $K/\mathbb{Q}$, is a fascinating subject, for cryptographic purposes we are interested in the size and structure over finite fields $\mathbb{F}_q$, with $q = p^k$. So, for now, we study $E(\mathbb{F}_q)$.

How large is $E(\mathbb{F}_q)$, roughly? Taking the short Weierstrass model as an example, we know there are only q^2 pairs $(x, y) \in \mathbb{F}_q \times \mathbb{F}_q$, and so at most we may get $q^2 + 1$ points (including the ‘point at infinity’). But this is a rather crude and frankly ridiculous upper-bound. In reality, given any $x_0 \in \mathbb{F}_q$, the probability that $x_0^3 + ax_0 + b$ is a square should essentially be fifty-fifty, and so, roughly half of the q possible x_0 values should give us *two* corresponding points $(x_0, \pm \sqrt{x_0^3 + ax_0 + b}) \in E_{a,b}(\mathbb{F}_q)$. Adding the point at infinity $0_E \in E(\mathbb{F}_q)$, which we always have, we should expect $|E(\mathbb{F}_q)|$ to be ‘close’ to $q + 1$.

Definition 4. For an elliptic curve E over $\mathbb{F}_q$, the *trace* t_q is the value

$$t_q := q + 1 - |E(\mathbb{F}_q)|. \quad (4)$$

◦

We will see later (in ??) why the name *trace* makes sense. For now, we may think of the trace as a measurement of how much larger or smaller $E(\mathbb{F}_q)$ is compared to the expected value of $q + 1$. It turns out that we have tight bounds on the possible sizes of $E(\mathbb{F}_q)$, or equivalently, the possible values of t_q , coming from ‘deep mathematics’ **TODO: cite Hasse and Silverman**.

Theorem 5. Let E be an elliptic curve over $\mathbb{F}_q$. Then $E(\mathbb{F}_q)$ satisfies

$$q + 1 - 2\sqrt{q} \leq |E(\mathbb{F}_q)| \leq q + 1 + 2\sqrt{q}. \quad (5)$$

We call this interval the *Hasse interval*. Equivalently, we get $|t_q| \leq 2\sqrt{q}$. ◦

If you are not amazed by this theorem the first time you see it, reflect on this result for a while. The result is extremely precise, giving only a narrow range

of options for the size of $E(\mathbb{F}_q)$ relative to q , in a very simple formula. As we will see later, the result is optimal as the bounds are ‘tight’, as we can show that elliptic curves exist for every integer in the Hasse interval, up to very few very particular exceptions.

SPECIAL VALUES. A few values for $|E(\mathbb{F}_q)|$, resp. t_q , are special. Clearly the value $|E(\mathbb{F}_q)| = q + 1$, resp. $t_q = 0$ is special, as it is precisely in the middle of the Hasse interval and matches with the ‘rough expectation’ sketched before. Two other special values are the bounds $q + 1 \pm 2\sqrt{q}$. Curves that reach such a bound are called *minimal curves*, resp. *maximal curves*. As a leading example, take $q = p^2$, so that the extremeties of the Hasse interval correspond to $t_{p^2} = \pm 2p$. It turns out that such curves display very special behaviour in a way that we can make precise later. From the fact that curves with trace $-2p, 0$ and $2p$ all display special behaviour, we may guess that any trace t_q such that $t_q \equiv 0 \pmod{p}$ is special. Indeed, this is the case, but we will only find out how special later on. For now, we simply define these values as ‘special’. **TODO: rewrite this last part**

Definition 6. Let E be an elliptic curve over $\mathbb{F}_q$. If $t_q \equiv 0 \pmod{p}$, then E is a *supersingular¹ elliptic curve*. Otherwise, E is an *ordinary elliptic curve*. $\circ$

Supersingular elliptic curves will be our core focus, as they are ‘essentially’ the only curves we use nowadays in isogeny-based cryptography. This is in large contrast with classical elliptic-curve cryptography, where we only use ordinary curves! We discuss this in a bit more detail in [Section 2](#). The following result is very helpful when working with supersingular elliptic curves.

Lemma 7. Let E be a supersingular elliptic curve over $\mathbb{F}_q$. Then E is isomorphic to a curve defined over $\mathbb{F}_{p^2}$.

Proof. We show this later, in ??, when we have developed more machinery. $\square$

Again, this result should surprise you, as it is *a priori* very mysterious why we should be able to represent such curves over $\mathbb{F}_{p^2}$. Furthermore, this result holds more generally for a supersingular curve over *any* field of positive characteristic, not just finite fields. Because of [Definition 7](#), let us more closely analyse the supersingular curves for the specific cases $q = p$ and $q = p^2$.

¹ The naming of these elliptic curves as *supersingular* is a historic blunder: it has confused students for decades as it has nothing to do with the usage of *singular* that we have used before. One should simply think of such curves as ‘special’ and if we could go back in time, we should have originally defined these curves as *special curves* instead of *supersingular*. In fact, when we later define what *superspecial* means, we will mean something even more special than *supersingular*. Alas, we are stuck with this name.

Lemma 8. Let E be a supersingular elliptic curve over $\mathbb{F}_p$ with $p \geq 5$. Then $|E(\mathbb{F}_p)| = p + 1$.

Proof. This is now easy: by [Definition 5](#), we have $t_p \leq 2\sqrt{p}$. If $p \geq 5$, then $t_p = 0$ is the only multiple of p smaller than $2\sqrt{p}$. $\square$

TODO: write on ss curves over $\mathbb{F}_{p^2}$, then summarize with the existence theorem from Deuring.

1.4 THE STRUCTURE OF ELLIPTIC CURVES

TODO: write on group law, torsion etc.

1.5 CURVE MODELS

TODO: why different models and when

§2 ELLIPTIC CURVES (CRYPTOGRAPHY)

II

ISOGENIES

isogenies placeholder

III

PAIRINGS

pairings placeholder

IV

ABELIAN VARIETIES

abelian varieties placeholder



HIGHER-DIMENSIONAL ISOGENIES

We assume basic familiarity with the following material:

- [Chapter I:](#)
- [Chapter II: Elliptic Curves \(cryptography\)](#)

higher-dimensional isogenies

Part 2

ADVANCED TOPICS IN ISOGENY-BASED CRYPTOGRAPHY

ADVANCED TOPICS IN ISOGENY-BASED CRYPTOGRAPHY

This part describes advanced topics in isogeny-based cryptography, such as the Deuring correspondence, higher-dimensional isogenies, and generalized class group actions.

Part 3

CRYPTOGRAPHIC CONSTRUCTIONS

CRYPTOGRAPHIC CONSTRUCTIONS

This part describes the cryptographic constructions, such as SQIsign or CSIDH, that are built on top of [Parts 1](#) and [2](#).

